

## Appendix: Control Protocol

### 1. Communication settings

#### 1.1 Baud rate:

Baud rate: 19200 without parity.

#### 1.2 Integer to Byte Conversion:

| Byte 0    | Byte 1    |
|-----------|-----------|
| Val % 256 | Val / 256 |

#### 1.3 General command structure

|   | Field       | Length<br>(bytes) |                                                  |
|---|-------------|-------------------|--------------------------------------------------|
| 1 | Sync char   | 1                 | 0x24                                             |
| 2 | Command     | 1                 | Command ID (See table below)                     |
| 3 | Data Length | 2                 | N - Length of the data section (in bytes)        |
| 4 | Data        | N                 | According to the command                         |
| 5 | CS          | 1                 | Check sum CS = Sync char ^ ... ^ Data[N - 1] Ack |

### 2. Control protocol

| Command           | Command ID | Data Length | Data    | Description                                          | Response                               |
|-------------------|------------|-------------|---------|------------------------------------------------------|----------------------------------------|
| BIT               | 0x0C       | 0           | No data |                                                      | BIT Response<br>General<br>Acknowledge |
| Report parameters | 0x02       | 0           | No data | Returns lens position                                | General<br>Response                    |
| Zoom In           | 0x08       | 0           | No data | Zoom lens movement<br>for constant encoder<br>counts | General<br>Acknowledge                 |

|                       |      |   |         |                                                                                 |                     |
|-----------------------|------|---|---------|---------------------------------------------------------------------------------|---------------------|
| Zoom Out              | 0x09 | 0 | No data | Zoom lens movement for constant encoder counts                                  | General Acknowledge |
| Zoom In continuous    | 0x23 | 0 | No data | Moves zoom lens until “Zoom Stop” command is send. Maintains focus all the way. | General Acknowledge |
| Zoom Out continuous   | 0x24 | 0 | No data | Moves zoom lens until “Zoom Stop” command is send. Maintains focus all the way. | General Acknowledge |
| Zoom Stop             | 0x14 | 0 | No data | Stops continuous zoom command                                                   | General Acknowledge |
| Focus Near            | 0x0A | 0 | No data | Focus lens movement for constant encoder counts                                 | General Acknowledge |
| Focus Far             | 0x0B | 0 | No data | Focus lens movement for constant encoder counts                                 | General Acknowledge |
| Focus Near continuous | 0x21 | 0 | No data | Moves focus lens until “Focus Stop” command is send.                            | General Acknowledge |
| Focus Far continuous  | 0x22 | 0 | No data | Moves focus lens until “Focus Stop” command is send.                            | General Acknowledge |
| Focus Stop            | 0x1B | 0 | No data | Stops continuous focus command                                                  | General Acknowledge |

|      |      |   |         |                                                                                                                                       |                                                                |
|------|------|---|---------|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
|      |      |   |         |                                                                                                                                       |                                                                |
| FOV1 | 0x0E | 0 | No data | Moves the lens to FOV1 at predefined focus position, maintains focus all the way.                                                     | General<br>Response<br>End of Action                           |
| FOV2 | 0x0F | 0 | No data | Moves the lens to FOV2 at predefined focus position, maintains focus all the way.<br><br>May be not available in discreet FOV lenses. | General<br>Response<br>End of Action<br>(if command available) |
| FOV3 | 0x10 | 0 | No data | Moves the lens to FOV3 at predefined focus position, maintains focus all the way.<br><br>May be not available in discreet FOV lenses. | General<br>Response<br>End of Action<br>(if command available) |
| FOV4 | 0x11 | 0 | No data | Moves the lens to FOV4 at predefined focus position, maintains focus all the way.<br><br>May be not available in discreet FOV         | General<br>Response<br>End of Action<br>(if command available) |

|                         |      |   |                                                                                                         |                                                                                                                                   |                                                             |
|-------------------------|------|---|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
|                         |      |   |                                                                                                         | lenses.                                                                                                                           |                                                             |
| FOV5                    | 0x12 | 0 | No data                                                                                                 | Moves the lens to FOV5 at predefined focus position, maintains focus all the way.<br>May be not available in discreet FOV lenses. | General Response<br>End of Action<br>(if command available) |
|                         |      |   |                                                                                                         |                                                                                                                                   |                                                             |
| Go Manually to Position | 0x0D | 4 | 1-2 bytes:<br>Zoom Encoder value<br>(Max 32000)<br><br>3-4 bytes:<br>Focus Encoder value<br>(Max 32000) |                                                                                                                                   | General Acknowledge                                         |
| Open Shutter            | 0x37 | 0 | No data                                                                                                 | Open shutter                                                                                                                      | General Acknowledge                                         |
| Close Shutter           | 0x38 | 0 | No data                                                                                                 | Close shutter                                                                                                                     | General Acknowledge                                         |

|                           |      |   |                                                                |                                   |                       |
|---------------------------|------|---|----------------------------------------------------------------|-----------------------------------|-----------------------|
| Get Shutter Data          | 0x44 | 0 | No data                                                        | Returns shutter status<br>Shutter | Shutter Response      |
| Get focal length          | 0x42 | 0 | No data                                                        | Returns focal length              | Focal Length Response |
| Go To Focal Length        | 0x43 | 2 | 1-2 bytes:<br>Focal Length (Max value 32000)                   |                                   | General Acknowledge   |
| Set Speed                 | 0x1F | 2 | 1 byte:<br>Zoom Speed (1 – 7)<br>2byte:<br>Focus Speed (1 – 7) |                                   | General Acknowledge   |
| Get Measured Temperatures | 0x45 | 0 | No data                                                        | Returns measured temperature      | Temperature Response  |

### 3. Responses

"End of Action" response is the same as the "GeneralResponse", but the command id is 0x29. This response will be generated at the end of lens movement (after Go to FOV, Focus in/out/Stop commands, Zoom in/out/stop commands etc.). This response will come after the lens moved to a requested position and stopped.

In addition this response will be generated during BIT command.

#### 3.1 General Response:

|   | Field       | Length |                                                        |
|---|-------------|--------|--------------------------------------------------------|
| 1 | Sync char   | 1      | 0x24                                                   |
| 2 | Command     | 1      | Requested command                                      |
| 3 | Data Length | 2      | 6                                                      |
| 4 | Data        | 2      | Current zoom encoder value                             |
|   |             | 2      | Current focus encoder value                            |
|   |             | 1      | Temperature (+/-127 signed int)                        |
|   |             | 1      | 0x00                                                   |
| 5 | ACK         | 1      | 0xFE – Successful operation<br>0xFF – Operation failed |
| 6 | CS          | 1      | Check sum                                              |

#### 3.2 Full End of Action:

|   | Field       | Length |                                 |
|---|-------------|--------|---------------------------------|
| 1 | Sync char   | 1      | 0x24                            |
| 2 | Command     | 1      | Requested command               |
| 3 | Data Length | 2      | 6                               |
| 4 | Data        | 2      | Current zoom encoder value      |
|   |             | 2      | Current focus encoder value     |
|   |             | 1      | Temperature (+/-127 signed int) |
|   |             | 1      | Current FOV                     |

|   |     |   |                                                            |
|---|-----|---|------------------------------------------------------------|
|   |     |   | 0 – Not in any specific FOV<br><br>1 … 5 – FOV1 … FOV5     |
| 5 | ACK | 1 | 0xFE – Successful operation<br><br>0xFF – Operation failed |
| 6 | CS  | 1 | Check sum                                                  |

### 3.3 BIT Response:

|   | Field       | Length |                                                                                                                                                                                                                                                                                                                                    |
|---|-------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Sync char   | 1      | 0x24                                                                                                                                                                                                                                                                                                                               |
| 2 | Command     | 1      | 0x0C                                                                                                                                                                                                                                                                                                                               |
| 3 | Data Length | 2      | 2                                                                                                                                                                                                                                                                                                                                  |
| 4 | Data        |        | BIT Flags:<br><br>Bit 0: Zoom Motor Error<br><br>Bit 1: Focus Motor Error<br><br>Bit 2: Zoom Encoder Error<br><br>Bit 3: Focus Encoder Error<br><br>Bit 4: Communication Error<br><br>Bit 5: Thermistor Error<br><br>Bit 6: Shutter Error (if available)<br><br>Bit 7: Shutter Encoder Error (not available)<br><br>8– 15: Reserve |
| 5 | ACK         | 1      | 0xFE – Successful operation<br><br>0xFF – Operation failed                                                                                                                                                                                                                                                                         |
| 6 | CS          | 1      | Check sum                                                                                                                                                                                                                                                                                                                          |

### 3.4 Shutter Response:

|   | Field       | Length |      |
|---|-------------|--------|------|
| 1 | Sync char   | 1      | 0x24 |
| 2 | Command     | 1      | 0x44 |
| 3 | Data Length | 2      | 3    |

|   |      |   |                                                        |
|---|------|---|--------------------------------------------------------|
| 4 | Data | 2 | 0 - shutter Encoder not available                      |
|   |      | 1 | Shutter state:<br>0 – open<br>1 – close<br>2 – Unknown |
| 5 | ACK  | 1 | 0xFE – Successful operation<br>0xFF – Operation failed |
| 6 | CS   | 1 | Check sum                                              |

### 3.5 Focal Length Response:

|   | Field       | Length |                                                        |
|---|-------------|--------|--------------------------------------------------------|
| 1 | Sync char   | 1      | 0x24                                                   |
| 2 | Command     | 1      | 0x2A                                                   |
| 3 | Data Length | 2      | 2                                                      |
| 4 | Data        | 2      | Focal Length value                                     |
| 5 | ACK         | 1      | 0xFE – Successful operation<br>0xFF – Operation failed |
| 6 | CS          | 1      | Check sum                                              |

### 3.6 General Acknowledge:

|   | Field       | Length |                                                        |
|---|-------------|--------|--------------------------------------------------------|
| 1 | Sync char   | 1      | 0x24                                                   |
| 2 | Command     | 1      | Requested command                                      |
| 3 | Data Length | 2      | 0                                                      |
| 4 | ACK         | 1      | 0xFE – Successful operation<br>0xFF – Operation failed |
| 5 | CS          | 1      | Check sum                                              |

### 3.7 Temperatures Response:

|  | Field | Length |  |
|--|-------|--------|--|
|--|-------|--------|--|

|   |             |   |                                                        |
|---|-------------|---|--------------------------------------------------------|
| 1 | Sync char   | 1 | 0x24                                                   |
| 2 | Command     | 1 | 0x45                                                   |
| 3 | Data Length | 2 | 3                                                      |
| 4 | Data        | 1 | 0                                                      |
|   |             | 1 | 0                                                      |
|   |             | 1 | measured temperature                                   |
| 5 | ACK         | 1 | 0xFE – Successful operation<br>0xFF – Operation failed |
| 6 | CS          | 1 | Check sum                                              |